

in vitro stim of T cells by plate bound anti-CD3 and soluble anti-CD28 antibodies

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Materials: (all located in TC 4 degree...Ab in box, binding buffer is in a filter flask)

-Bicarbonate Binding buffer, pH9.6

-Goat anti hamster antibody, Vector Laboratories, Cat# AI-9100

Follow instructions for resuspension (add 500ul H₂O, NOT PBS) This makes a 3mg/ml solution. It contains azide. Comes in a small brown bottle with a grey rubber lid.

-anti-CD3 (clone 2C11) we have a large amount from UCSF Ab core @ 1.8mg/ml (CONTAINS NO AZIDE...KEEP STERILE!)

-anti-CD28 (clone 37.51) we have a large amount from UCSF Ab core @ 1.0mg/ml (CONTAINS NO AZIDE...KEEP STERILE!)

Procedure:

Coat wells with 1/100 goat anti hamster antibody in binding buffer. Place in the 37 degree TC incubator for 1 hour. Suggested volumes sufficient to cover the bottom of the well yet not waste antibody are listed in the table below.

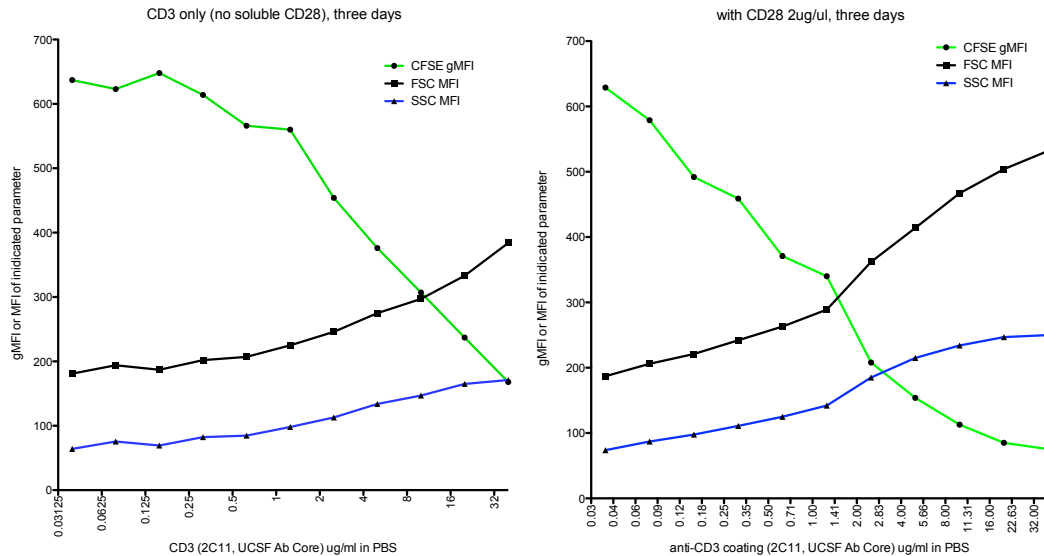
	Coating Volumes	
96well	50ul	After the 1 hour at 37 degrees, aspirate and rinse in PBS (use about ~2X the volume you used to coat...the bicarbonate buffer contains sodium azide, which you want to wash out)
48well	100ul	Aspirate PBS (be careful not to let plate dry after aspiration)
24well	250ul	Add anti-CD3 (2C11) diluted in PBS and incubate 1 hour at 37 degrees
12well	500ul	
6well	1ml	Aspirate and rinse in PBS (with ~2X the volume you used to coat)

Dilute cells in media (1 million per ml is a good density) and add 1ug/ml soluble CD28 Ab if costimulation is desired. Add cells+media to coated wells. Of course have an uncoated well for a negative control.

T cell density in your culture will affect proliferation. So unpurified splenocytes plated at 1 million per ml will proliferate less than purified T cells at the same density. However, too dense and your culture will go acidic (yellow) before the end of your assay (typically 3 days) which of course will negatively affect proliferation.

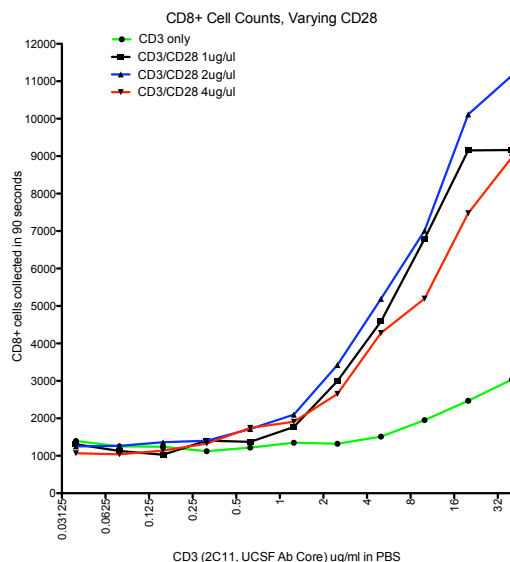
For simplicity, I plate unpurified or purified cells @ 1million/ml., in spite of the issues listed above.

In a trial experiment, the CD8+ gate of unpurified splenocytes at 1 million per ml using the above protocol proliferated as follows:



So based on this, for maximal stimulation, I would suggest a anti-CD3 coating dose of 20ug/ml (or a 1/90 dilution of our 1.8mg/ml stock). An anti-CD3 low dose could be 1ug/ml, or 1/1800 of our anti-CD3 stock in PBS.

1ug/ml soluble CD28 is sufficient for maximal co-stimulation (see below). Diluting goat anti-hamster below 1/100 reduces proliferation for all but the highest CD3 doses (@20ug/ml...not shown). **For co-stimulation, add 1/1000 anti-CD28 to the cells+media just before or after plating.**



NOTE: As mentioned above, denser cultures or longer coating with goat anti hamster and/or CD3 may shift these values slightly. For example, I have done 2 hour coating with the 1/100 gAH and then 2h with commercial (eBio) CD3 (2C11) antibody followed by 1 million per ml of purified T cells or 2.5 million per ml of splenocytes. Using this protocol/reagents, I found the CD3 max stimulation dose to be 5ug/ml and an appropriate CD3 lo dose to be around 0.6-0.7 ug/ml.